Discussion section 4

Huangchen Zhou

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1 Chp 3, Q23

Consider a k-state Markov chain with transition matrix

$$P = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ \vdots \\ k-2 \\ k-1 \\ k \end{array} \begin{array}{c} \begin{array}{ccccccc} 1 & 2 & 3 & \cdots & k-2 & k-1 & k \end{array} \\ \begin{pmatrix} 1/k & 1/k & 1/k & \cdots & 1/k & 1/k & 1/k \\ 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & \cdots & 1 & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 0 \end{pmatrix} \end{array}.$$

Show that the chain is ergodic and find the limiting distribution.

Proof. Since the Markov chain is finite, it suffices to check the chain is irreducible and aperiodic.

Irreducible: For any $i < j$, one can reach from i to j by $i \to 1 \to j$; and one can reach from j to i by $j \to j-1 \to j-2 \to \cdots \to i+1 \to i$.

Aperiodic: It suffices to check state 1 has period one, which is clear since $P_{11} = 1/k > 0$.

By fundamental theorem for ergodic Markov chains(cf. Theorem 3.8 in the textbook), the limiting distribution is just the stationary distribution π of P.

Say $\pi = (\pi_1, \dots, \pi_k)$. Then $\pi P = \pi$ becomes

$$\begin{cases} \frac{1}{k}\pi_1 + \pi_2 = \pi_1, \\ \frac{1}{k}\pi_1 + \pi_3 = \pi_2, \\ \vdots \\ \frac{1}{k}\pi_1 + \pi_{k-1} = \pi_{k-2}, \\ \frac{1}{k}\pi_1 + \pi_k = \pi_{k-1}, \\ \frac{1}{k}\pi_1 = \pi_k. \end{cases}$$

Thus $\pi_j = (1 - \frac{j-1}{k})\pi_1$ for any $1 \le j \le k$. Together with the normalizing condition we have

$$1 = \sum_{j=1}^{k} \pi_j = \sum_{j=1}^{k} (1 - \frac{j-1}{k})\pi_1 = (k - \frac{k-1}{2})\pi_1 = \frac{k+1}{2}\pi_1.$$

So $\pi_j = (1 - \frac{j-1}{k}) \cdot \frac{2}{k+1}$ for any $1 \le j \le k$.

$\square$

2 Chp 3, Q26

Assume that $(p_1, \dots, p_k)$ is a probability vector. Let P be a $k \times k$ transition matrix defined by

$$P_{ij} = \begin{cases} p_j & \text{if } i = 1, \dots, k-1, \\ 0 & \text{if } i = k, j < k, \\ 1 & \text{if } i = k, j = k. \end{cases}$$

Describe all the stationary distributions for P.

Proof.

$$P = \begin{bmatrix} p_1 & p_2 & p_3 & \cdots & p_{k-2} & p_{k-1} & p_k \\ p_1 & p_2 & p_3 & \cdots & p_{k-2} & p_{k-1} & p_k \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ p_1 & p_2 & p_3 & \cdots & p_{k-2} & p_{k-1} & p_k \\ p_1 & p_2 & p_3 & \cdots & p_{k-2} & p_{k-1} & p_k \\ 0 & 0 & 0 & \cdots & 0 & 0 & 1 \end{bmatrix}.$$

Say $\pi = (\pi_1, \ldots, \pi_k)$. Then $\pi P = \pi$ becomes

$$\begin{cases} \pi_1 p_1 + \cdots + \pi_{k-1} p_1 = \pi_1, \\ \pi_1 p_2 + \cdots + \pi_{k-1} p_2 = \pi_2, \\ \vdots \\ \pi_1 p_{k-2} + \cdots + \pi_{k-1} p_{k-2} = \pi_{k-2}, \\ \pi_1 p_{k-1} + \cdots + \pi_{k-1} p_{k-1} = \pi_{k-1}, \\ \pi_1 p_k + \cdots + \pi_{k-1} p_k + \pi_k = \pi_k. \end{cases}$$

Case 1: If $p_k \neq 0$, then the last equation in the system reads $\pi_1 + \cdots + \pi_{k-1} = 0$. Together with the other equations in the system we have $\pi_1 = \pi_2 = \cdots = \pi_{k-1} = 0$. Hence $\pi = (0, 0, \cdots, 0, 0, 1)$ if $p_k \neq 0$.

Case 2: If $p_k = 0$, then $\pi_j = (1 - \pi_k) p_j$ for each $1 \leq j \leq k-1$ (For example, the first equation in the system reads, $(\pi_1 + \cdots \pi_{k-1}) p_1 = \pi_1$, i.e.$(1 - \pi_k) p_1 = \pi_1$). Hence by relabeling π_k by a,

$$\pi = ((1-a)p_1, (1-a)p_2, \cdots, (1-a)p_{k-2}, (1-a)p_{k-1}, a)$$

if $p_k \neq 0$.

$\square$

3 Chp 3, Q29

Consider a Markov chain with transition matrix

$$P = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \\ g \end{array} \begin{array}{c} \begin{array}{ccccccc} a & b & c & d & e & f & g \end{array} \\ \begin{pmatrix} 0.6 & 0.2 & 0 & 0 & 0.2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.5 & 0.5 \\ 0 & 0 & 0.3 & 0.7 & 0 & 0 & 0 \\ 0 & 0.3 & 0.4 & 0.3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0.1 & 0 & 0 & 0 & 0 & 0.9 \\ 0 & 0.2 & 0 & 0 & 0 & 0.8 & 0 \end{pmatrix} \end{array}.$$

Identity the communication classes. Classify the states as recurrent or transient, and determine the period of each state.

Proof. With a graph in hand one can identity the communication classes:

1. $\{a\}$, transient, with period 1.
2. $\{e\}$, recurrent, with period 1.

3. $\{c, d\}$, transient, with period 1.
4. $\{b, f, g\}$, recurrent, with period 2.

$\square$